Lab guide – Gateway redundancy and load balancing

1 Objectives

The aim of this laboratory work is to study gateway redundancy and load balancing using the HSRP and GLBP protocols.

2 Preparation

See the slides "Gateway redundancy and load balancing".

3 Equipment

These exercises are performed using GNS3 VM, equipped with Docker attacker and Cisco 7200 routers. Use Cisco 3725 routers as PCs.

4 HSRP

The goal of this exercise is to understand the operation of the HSRP protocol. Consider the network of Figure 1. All routers must be 7200 Cisco routers. Use OSPF as a routing protocol but configure the interfaces f0/0 of routers R1, R2, and R3 as passive interfaces. The loopback address of R4 represents an Internet destination.

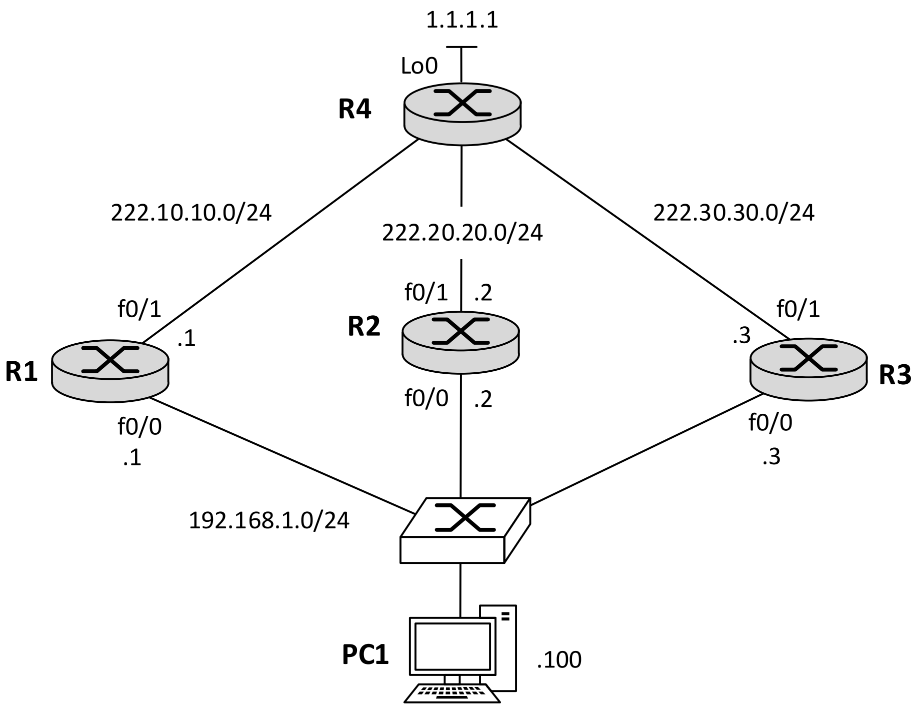

Figure 1: HSRP and GLBP.

The configuration of HSRP at R1 is the following:

```
int f0/0
standby 1 ip 192.168.1.254
standby 1 priority 150
standby 1 preempt
```

The two last instructions are optional; the default priority is 100. Useful show commands are **show standby** and **show standby brief**. Also useful may be the **debug standby events** command.

Summary of what is to be explained:

1. Register the physical MAC addresses of the R1, R2, R3, and PC1 interfaces (at 192.168.1.0/24). Use the **show standby** command to identify the MAC address of the Virtual gateway. What is the rule for defining the MAC address of a Virtual gateway?
2. When PC1 pings the Internet destination what is the destination MAC address of the ICMP Echo Request message?
3. What are the source MAC addresses of the HSRP Hello messages sent by the gateways? Based on this, explain how the switch learns the MAC address used by the Active gateway. What changes at the switch when the Active gateway changes?
4. What is the role of the Active, Standby, and Listen gateways? What is the criterion for defining which gateway has a given role? How is this influenced by the priority and preempt features? Illustrate your findings using experimentation, e.g., shutting down, changing the priority, or activating the preemption of the gateway's interfaces.
5. What are the protocol actions when the Standby gateway is preempted to become the Active gateway? To answer this question read the specification of HSRP (RFC 2281) and analyze the HSRP messages using Wireshark.

5 HSRP with object tracking

You will now study the object tracking feature of HSRP. There are two variants: interface tracking and IP SLA tracking.

The configuration of the HSRP interface tracking alternative at all routers is the following:

```
track 1 interface f0/1 line-protocol
exit
int f0/0
standby 1 ip 192.168.1.254
standby 1 preempt
standby 1 track 1 decrement 50
```

The configuration of the HSRP IP SLA tracking alternative at R1 is the following:

```
ip sla 1
icmp-echo 1.1.1.1
frequency 10
```

```
exit
ip sla schedule 1 start-time now life forever
track 1 ip sla 1
int f0/0
standby 1 ip 192.168.1.254
standby 1 preempt
standby 1 track 1 decrement 50
```

A useful show command is **show ip sla statistics**.

Summary of what is to be explained:

1. Before doing the object tracking configurations, identify the Active gateway, shut down its f1/0 interface and ping from PC1 to the Internet. Explain the result.
2. With the interface tracking alternative configured, shutdown the f1/0 interface of the Active gateway, and ping from PC1 to the Internet destination. Repeat this procedure until all f1/0 interfaces are shutdown. Explain the result.
3. With the IP SLA tracking alternative configured, observe the ICMP packets sent from R1, R2, and R3, to the Internet destination? What is their role?
4. Now repeat the procedure of the interface tracking alternative. Are the results different?
5. When should each alternative be used?

6 Attacking HSRP

You may already be thinking on how to attack HSRP! Include a Docker attacker in the internal network and write a Scapy script to make the attacker spoofing the Active gateway. Then use authentication of HSRP messages to mitigate the attack. The configuration of authentication is done in the following way:

```
int f0/0
standby 1 authentication md5 key-string MY_SECRET_KEY
```

Summary of what is to be explained:

1. Explain how you performed the attack, and how it was mitigated. Is this type of authentication secure?

7 GLBP

The goal of this exercise is to understand the operation of GLBP. Use the network of Figure 1, but with four PCs instead of one. The configuration of GLBP at R1 is the following:

```
int f0/0
glbp 1 ip 192.168.1.254
glbp 1 priority 150
glbp 1 preempt
```

The two last instructions are optional. Useful show commands are **show glbp** and **show glbp brief**.

Summary of what is to be explained:

1. Using the **show glbp** command identify the physical and virtual MAC addresses of each GLBP interface, and the AVG and the AVFs. Does each GLBP router know the virtual MAC addresses of the others? What is the rule for defining the virtual MAC addresses?
2. Analyze the GLBP messages transmitted by each router. Which type of source MAC address is being used to send the GLBP messages?
3. Analyze the ARP messages at the AVG interface when pinging from each PC to the Internet destination. Which router responds to ARP Requests? What type of MAC address does it provide to the requesting PCs? What is the gateway being used by each PC? What is the load balancing method being used and how does it work?
4. What are the criteria for defining the Active, Standby, and Listen AVGs? How is this influenced by the priority and preempt features? Is this different from HSRP?
5. What happens when one GLBP router fails? Can PCs using the failed PC still communicate with the Internet destination? How?
6. Can you give a possible explanation on why GLBP uses virtual MAC addresses and not the actual physical MAC addresses?

8 GLBP with weighted load balancing

In this experiment you will address GLBP with weighted load balancing. Use only two gateways, i.e., R1 and R2, and the following configuration at R1:

```
int f0/0
glbp 1 ip 192.168.1.254
glbp 1 load-balancing weighted
glbp 1 weighting 100
```

The configuration of R2 is the same except that the weight must be 50.

Summary of what is to be explained:

1. Analyze the ARP messages at the AVG interface and the output of the **show glbp** command (in particular, the client selection count of each forwarder). Which PCs were assigned to each gateway? Is this in agreement with weights assigned to the gateways?